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Quinn Booth

github.com/quinnbooth

Education

Columbia University | *The Fu Foundation School of Engineering and Applied Science* Expected Graduation 05/2024

- B.S. in Electrical Engineering with a minor in Computer Science
- Relevant Coursework: User Interface Design, Data Structures, Advanced Programming, Databases, Cloud Computing with AWS, Computer Networks, Applied ML, Empirical Methods of Data Science, Advanced Logic Design, Embedded Systems
- GPA: 3.92 / 4.0

Skills

Languages: Python, JavaScript, TypeScript, HTML, CSS, SQL, C++, C, Verilog, SystemVerilog, Java.
Technical: React, Node, Flask, AWS, GCP, REST API, GraphQL, MySQL, PyTorch, sklearn, Linux/Unix, Scripting, Git.
Leadership: Head of 3 independent research teams, mentored 40+ students in data science and machine learning.
Communication: Journalism experience with a personal mentor, fluent English and intermediate Spanish.

Research Experience

Research Assistant | Internet Real-Time Lab | Dept. of Computer Science, Columbia University 09/2023 - Present

- Developed firmware in C++ for low power IoT devices, improving energy efficiency by >10 times. Designed and power profiled LoRa communication circuitry. Project culminates in a co-authored conference paper with Dr. Henning Schulzrinne.

Research Assistant | Creative Machines Lab | Dept. of Computer Science, Columbia University 09/2022 - Present

- Led software/firmware design for the Stochastic Particle Robots project, using REST APIs in C++ to send a fleet of robots instructions over the cloud. Conference paper accepted to IEEE ReMAR 2024. Designed the 16+ robots' circuitry and PCBs.

Research Assistant | Creative Machines Lab | Dept. of Computer Science, Columbia University 05/2022 - Present

- Managed the Artificial Metabolism project's hardware team, prototyping and using computer vision in Python to track robotic motion. Co-authored two journal/conference papers with Dr. Hod Lipson. Work featured in upcoming Bloomberg documentary.

Research Assistant | Dept. of Statistics, Columbia University 07/2023 - 10/2023

- Implemented cutting-edge machine learning algorithms in Python, using rate-distortion-perception theory to enhance media compression with Dr. Marcel Nutz. Completed an extensive literature review (20+ papers) on modern optimization techniques.

Work Experience & Projects

AWS Social Media Site: Created a React app with GPT2 integration, hosted using AWS Amplify, DynamoDB, S3 and containers.
Bachelor Nation Experience: Used Flask, jQuery and Bootstrap to produce an interactive app, teaching how The Bachelor works.
Haunts of Manhattan: Built a site cataloging popular haunted sites in Manhattan, utilizing a Flask backend database.
Fractal Visualizer: Developed an interactive website, featuring various fractals such as the Mandelbrot set and Barnsley fern.

Teaching Assistant | Dept. of Computer Science, Columbia University 08/2023 - Present

- Worked with Dr. Michelle Levine on graduate-level course Empirical Methods of Data Science. Prepared live demos on deep learning and statistical data analysis, using sklearn and PyTorch. Assisted the writing and grading of assignments in Python.

STAR Scholar | Dept. of Statistics, Columbia University 04/2023 - 07/2023

- Trained intensively in machine learning, data science, statistics and probability at the Summer Training for Aspiring Researchers program supervised by Dr. Banu Baydil.

News Editorial Intern | The Taiwan Times | Kaohsiung City, Taiwan (remote) 05/2022 - 08/2022

- Published 12 news and opinion pieces on The Taiwan Times' official website, covering modern scientific discoveries. Cultivated strong writing and communication skills under a personal industry mentor, Hedy Heng Yang.

Tech I (Engineer) | The Escape Game | Nashville, TN 12/2018 - 11/2019

- Produced software in C++ and designed circuitry for puzzles in escape rooms across the U.S. Managed warehouse inventory.